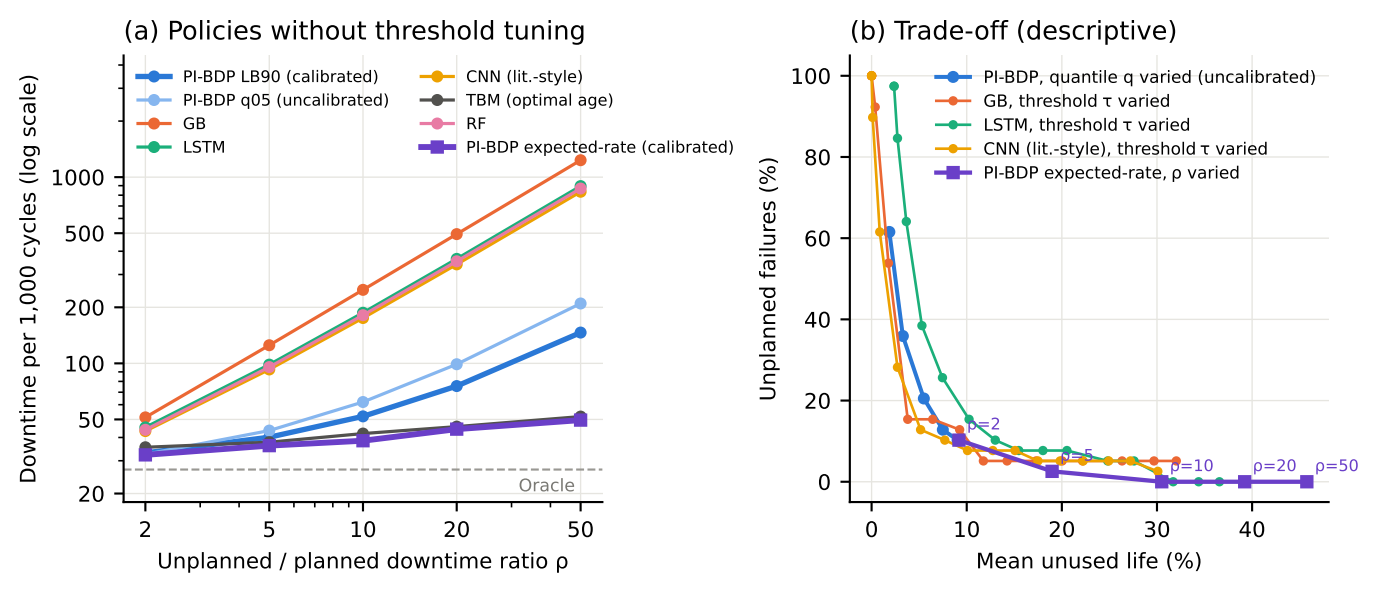
(a) Policies without threshold tuning
PI-BDP LB90 (calibrated)
PI-BDP q05 (uncalibrated)
GB
LSTM
CNN (lit.-style)
TBM (optimal age)
RF
PI-BDP expected-rate (calibrated)
Downtime per 1,000 cycles (log scale)
1000
500
200
100
50
20
Oracle
2
5
10
20
50
Unplanned / planned downtime ratio ρ
(b) Trade-off (descriptive)
PI-BDP, quantile q varied (uncalibrated)
GB, threshold τ varied
LSTM, threshold τ varied
CNN (lit.-style), threshold τ varied
PI-BDP expected-rate, ρ varied
Unplanned failures (%)
100
80
60
40
20
0
ρ=2
ρ=5
ρ=10
ρ=20
ρ=50
0
10
20
30
40
Mean unused life (%)